Exercise: 33

Paper chromatography experiment

Aim:

To separate and study the photosynthetic pigments (chloroplast pigments) by paper chromatography method.

Requirements:

Fresh spinach leaves, chromatography paper (whatman No.1), a wide long test tube, a split cork, mortar & pestle, petroleum ether, acetone, funnel, beaker, filter paper, capillary tube, sand etc.,

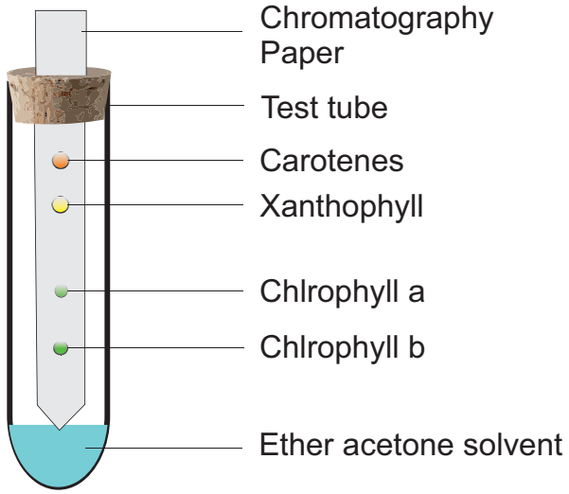


Figure 29 : Paper chromatography experiment

Procedure:

1. Grind a few spinach leaves with little fine sand and about 5 ml of acetone in a mortar and pestle. Filter it to get acetone extract of the leaf pigments.
2. Take a narrow strip of chromatographic paper (Whatman No.1). Cut one end of the strip into a narrow notch.
3. Put a drop of the pigment extract in the middle of the strip near the notch with the help of capillary tube. Allow the drop to dry and repeat till four or five drops are placed on the paper.
4. Take the test tube and pour about 5 ml of ether acetone solvent (9 ether : 1 acetone) in it. Now hang the pigment extract loaded chromatographic strip in the test tube with the help of a split cork, in such a way that the loading spot lies about 1 cm above the solvent level.
5. Make the cork air tight and place the test tube undisturbed for some time, when solvent rises about 3/4th of the strip, take out the strip carefully and let it dry.

Observation:

After one hour observe the chromatographic paper. The Photosynthetic pigments being separated into four distinct bands. Different leaf pigments can be identified by their colours.

Carotene	Xanthophyll	Chlorophyll a	Chlorophyll b
Yellow Orange	Yellow	Bluish Green	Greenish Yellow

Inference:

Photosynthetic pigments chlorophyll b, chlorophyll a, xanthophyll and carotenes are separated on the chromatographic paper. Presence of different photosynthetic pigments in chloroplast is proved.

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